Master Thesis

Modelling Latin American Corporate Bond Risk:
A Regime-Aware Framework for Z-Spread Risk Decomposition

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Prefatory Note

This document constitutes my **Master thesis**, completed at **Paris 1 Panthéon-Sorbonne** as part of the Master Finance Technology Data programme.

The thesis focuses on the **modelling of Z-spread risk in USD-denominated Latin American corporate bonds**. Its objective is to develop a regime-aware framework capable of explaining corporate bond spreads through several complementary layers of risk, including **bond-level characteristics**, **issuer and sector fundamentals**, **macro-financial conditions**, and **emerging-market-specific factors**, using both **standard** and **alternative data**.

The empirical universe is drawn from the **Latin American corporate segment of the Bloomberg EM USD Aggregate Total Return Index Value Unhedged**, restricted to issuers whose economic country of risk belongs to Latin America excluding Caribbean territories. The analysis focuses on **standard fixed-coupon corporate bonds**, with spread dynamics modelled as a function of conventional credit, liquidity, macro-financial, and Latin American risk channels.

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1 Introduction and Research Framework

1.1 Motivation

This thesis contributes to the analysis of **emerging-market corporate bonds**, with a deliberate focus on the **Latin American sub-universe**. The broader emerging-market universe encompasses an extremely heterogeneous set of economies — from advanced commodity exporters in the Gulf to technology-driven exporters in Asia, from large diversified economies such as China and India to smaller frontier markets — each driven by distinct macroeconomic regimes, monetary policy frameworks, commodity cycles, and political dynamics. Pooling such heterogeneous issuers into a single empirical model risks masking the specific risk channels that matter for each region and producing decompositions that are difficult to interpret economically.

Latin America represents a particularly coherent and analytically tractable sub-universe. The region shares a set of structural features — commodity dependence, U.S. dollar sensitivity, common exposure to global risk appetite, regional contagion dynamics, and political-cycle-driven volatility — that make it amenable to a unified risk-decomposition framework. At the same time, it encompasses a rich cross-section of countries (Mexico, Brazil, Colombia, Chile, Peru, Costa Rica, Guatemala, and others), sectors (energy, financials, utilities, industrials, consumer), and credit qualities, providing sufficient dispersion to identify both common and idiosyncratic sources of spread variation.

Focusing on Latin America also allows the empirical design to be sharper: the emerging-market-specific variables can be tailored to the region — incorporating LATAM CDS composites, regional FX dynamics, and country-level geopolitical risk indices — rather than relying on broad pan-EM proxies that may not capture the transmission mechanisms relevant to individual Latin American issuers.

Observed spreads in this segment are shaped by multiple layers of risk — issuer fundamentals, bond-level features, macro-financial conditions, and Latin American vulnerabilities — whose relative importance may shift materially across market environments. In calm periods, spreads may be driven primarily by issuer quality, maturity, liquidity, and sector characteristics. In stress periods, however, global funding conditions, U.S. rates, dollar strength, sovereign risk, and regional contagion may become dominant sources of repricing.

The thesis adopts a **risk-modelling perspective** based on the **Z-spread**, used as a consistent bond-level measure of compensation above the benchmark curve. It proposes a **regime-aware and economically structured framework** to assess how spread determinants evolve across market states and how Latin American corporate bond spreads can be decomposed into macro, emerging-market, bond-specific, and latent common components.

1.2 Research Framework

The central research question is:

To what extent can a regime-aware and economically structured model, combining standard and alternative data, explain the risk of Latin American corporate bonds through the variation of their Z-spreads?

The analysis is organized around three complementary dimensions: **bond-level factors**, **macro-financial factors**, and **Latin American risk factors**. A regime-identification layer captures changes in market conditions and the possibility that the sensitivity of spreads to these factors varies across states of the world. In stressed environments, the balance between credit protection, market liquidity, and investor risk appetite may be materially reassessed.

1.3 Scope and Target Variable

The empirical analysis draws on the **Latin American corporate segment of the Bloomberg EM USD Aggregate Total Return Index Value Unhedged**, restricted to issuers whose `CNTRY_OF_RISK` belongs to the 17 Latin American economies excluding Caribbean territories. The thesis focuses on **standard fixed-coupon corporate issuers**, excluding sovereign, quasi-sovereign, and government-sector bonds.

The target variable is the **Z-spread**, denoted $z_{i,t}$ for bond i at time t, defined as the constant spread added to the benchmark zero-coupon curve so that discounted cash flows equal the market price. It serves as a consistent **bond-level measure of risk compensation**, capturing how markets price the combined effects of issuer fundamentals, bond structure, liquidity conditions, macro-financial variables, and Latin American exposure.

2 Literature Review

The literature reviewed here highlights four main points: issuer fundamentals matter, but they do not fully explain bond spreads; common market and macro-financial factors drive a large share of spread variation; non-default components such as liquidity and systematic risk premia are economically important; and spread behaviour is cyclical, varying across market and business-cycle conditions. In Latin America, these effects are reinforced by the interaction between domestic conditions, external spillovers, commodity cycles, and regional common factors. For the Latin American corporate bond universe considered here, spreads must therefore be analysed as the joint outcome of issuer risk, liquidity, cyclical macro-financial conditions, and broader market-wide pricing forces.

2.1 Default Risk, the Credit Spread Puzzle, and Non-Default Components

Structural models of credit risk predict that bond spreads increase with leverage, asset volatility, and declining distance to default. In this framework, spreads compensate investors for expected losses and default risk.

A direct quantitative decomposition of corporate–Treasury spread *levels* is provided by Elton et al. (2001). Using dealer-quoted U.S. investment-grade bonds, they argue that spreads should be studied using *spot-rate* differentials rather than yield-to-maturity differentials, and decompose corporate spot spreads into compensation for expected default losses, a tax component, and a residual component attributed to systematic risk premia. Their central empirical result is that expected default losses account for only a small fraction of observed spreads, while the remaining portion is economically large and varies systematically with priced market factors.

While Elton et al. (2001) focus on spread *levels*, Collin-Dufresne, Goldstein, and Martin (2001) study the determinants of *monthly changes* in credit spreads using a large panel of noncallable industrial bonds. They find that standard structural variables have limited explanatory power for spread changes and that a dominant common factor remains in residual spread innovations. This suggests that spread dynamics are strongly driven by market-wide repricing forces rather than purely issuer-specific fundamentals.

This wedge between spreads and expected default losses is commonly referred to as the "credit spread puzzle." Amato and Remolona (2003) document that observed spreads often materially exceed actuarial expected losses and argue that this gap reflects compensation for undiversified downside risk, liquidity premia, taxes, and systematic risk compensation rather than a simple anomaly.

Importantly, Feldhütter and Schaefer (2018) show that part of the apparent puzzle may reflect imprecise estimation of default probabilities, especially when based on short historical samples. Their results imply that the expected-default anchor itself is estimated with uncertainty, which is relevant for any fair-value spread framework.

Liquidity provides an economically important non-default component in bond markets. Using a broad panel of U.S. corporate bonds, Chen, Lesmond, and Wei (2007) document that illiquidity is significantly priced in yield spreads using bid–ask spreads, zero-return measures, and transaction-cost proxies. This motivates treating liquidity as an explicit state variable in spread modelling rather than absorbing it into residual noise.

Measurement considerations further reinforce the importance of market-implied variables. Jacobs and Karagozoglu (2010) show that CDS spreads adjust more rapidly than agency ratings and embed forward-looking information about credit risk. This motivates incorporating market-based indicators alongside issuer or macro variables whenever possible.

Taken together, this literature shows that observed bond spreads are not reducible to expected default losses. A large share reflects time-varying compensation for systematic risk and liquidity frictions (Feldhütter and Schaefer 2018). This decomposition logic is directly useful for the Latin American corporate bond universe considered here, where spreads embed both firm-level credit risk and broader regional risk premia.

2.2 Emerging Markets, Common Factors, and Spread Determinants

The literature on emerging-market debt highlights that bond pricing in these markets reflects the joint effect of domestic fundamentals, global financial conditions, and strong common factors.

Erb, Harvey, and Viskanta (1999) document key stylized facts about emerging-market debt: high volatility, crisis-sensitive correlation patterns, and strong sensitivity to country risk. Their evidence implies that bond spreads should not be interpreted as purely issuer-specific signals; they are also exposed to common repricing forces, especially in stress periods. These stylized facts are particularly pronounced in Latin America, where external shocks, commodity price cycles, and political events have historically triggered sharp and synchronised spread movements.

Miyajima, Mohanty, and Chan (2012) study local-currency government bonds in 11 emerging economies and show that domestic factors—especially monetary policy and fiscal conditions—explain a substantial share of bond-yield dynamics, while external variables such as U.S. Treasury yields and global risk sentiment remain important. The core implication is directly relevant to the present thesis: Latin American corporate bond spreads are jointly driven by issuer conditions, domestic macroeconomic conditions, and global spillovers.

Díaz Weigel and Gemmill (2006) deepen this argument by showing that sovereign creditworthiness in emerging markets is driven mainly by common factors rather than purely domestic fundamentals. Using distance-to-default estimates from Brady bond prices, they find that regional factors explain the largest share of variation, followed by global conditions, while country-specific factors explain only a relatively small share. Their contagion-based interpretation of spread dynamics is directly applicable to Latin America, where regional transmission through investor flows, currency pressure, and commodity co-movement is well documented.

These findings imply that Latin American corporate bond spreads depend on a joint interplay of issuer fundamentals, country conditions, regional repricing, and external financial conditions. Corporate credit spreads therefore remain embedded in broader pricing dynamics shaped by global rates, risk sentiment, liquidity, and regional common factors.

2.3 Systematic Risk, Common Factors, and the Business Cycle

The strong common variation documented by Collin-Dufresne, Goldstein, and Martin (2001) implies that systematic pricing forces dominate short-horizon spread movements. This motivates modelling approaches that allow for latent market-wide spread factors rather than relying exclusively on observed covariates.

Elton et al. (2001) further demonstrate that the residual component of spreads covaries with priced market factors, providing direct evidence that bond spreads embed systematic risk premia.

A macro-financial interpretation of these systematic forces is provided by Gilchrist and Zakrajšek (2012). Using a large panel of corporate bond prices, they construct an aggregate credit spread index and decompose it into a default-related component and an *excess bond premium* (EBP), defined as the portion unexplained by firm-specific expected default risk. They show that the EBP is countercyclical and highly predictive of future real economic activity, supporting an interpretation in which spreads widen when intermediary risk-bearing capacity deteriorates.

For the Latin American corporate bond universe considered here, this logic extends naturally: spreads may widen not only because issuer fundamentals deteriorate, but because global financial conditions tighten, funding constraints increase, U.S. dollar liquidity weakens, or risk appetite toward Latin American credit declines. The evidence from Erb, Harvey, and Viskanta (1999), Miyajima, Mohanty, and Chan (2012), and Díaz Weigel and Gemmill (2006) suggests that common and external factors are essential for understanding spread dynamics in this universe.

Together, these findings motivate regime-aware modelling structures in which latent states capture shifts in market-wide spread premia, interpretable as changes in global risk appetite, credit supply, capital flows, and funding conditions (Gilchrist and Zakrajšek 2012).

Beyond macro-credit applications, regime-switching methods have been shown to improve asset-allocation and factor-investing performance when risk premia are state-dependent. Wang, Lin, and Mikhelson (2020) implement a Gaussian Hidden Markov Model (HMM) on equity-market return and volatility data to identify latent bull, bear, and intermediate regimes. Their methodological implication is directly relevant here: when premia are regime-dependent, conditioning spread models and valuation signals on latent states can materially enhance explanatory power and portfolio usefulness.

2.4 Implications for This Thesis

Taken together, the literature reviewed above leads to four implications for the empirical design of this thesis:

- **Default Risk, the Credit Spread Puzzle, and Non-Default Components:** default-related fundamentals provide an important anchor for spread analysis, but they are not sufficient to explain the full level of observed spreads. A substantial share reflects non-default components, especially liquidity premia and compensation for systematic risk (Elton et al. 2001; Collin-Dufresne, Goldstein, and Martin 2001; Feldhütter and Schaefer 2018; Chen, Lesmond, and Wei 2007; Amato and Remolona 2003).
- **Emerging Markets, Common Factors, and Spread Determinants:** in Latin America, bond spreads are shaped not only by issuer or sovereign-specific conditions, but also by external spillovers, FX risk, commodity cycles, and strong regional common factors (Miyajima, Mohanty, and Chan 2012; Díaz Weigel and Gemmill 2006; Erb, Harvey, and Viskanta 1999).
- **Systematic Risk, Common Factors, and the Business Cycle:** spread dynamics are strongly influenced by market-wide pricing forces and cyclical financial conditions, implying that spread behaviour varies across different market environments (Collin-Dufresne, Goldstein, and Martin 2001; Gilchrist and Zakrajšek 2012; Wang, Lin, and Mikhelson 2020).
- **Implications for This Thesis:** the analysis will place particular emphasis on **non-fundamental components** of spreads, especially liquidity, market-wide pricing forces, Latin American external conditions, FX risk, and latent cyclical dynamics, while treating issuer fundamentals as an important background anchor. The focus on Latin America allows the emerging-market risk channel to be measured with region-specific instruments — LATAM CDS composites, regional FX scores, and country-level geopolitical risk indices — rather than pan-EM proxies.

3 Methodology

This section presents the integrated empirical methodology, from the theoretical decomposition structure to the three operational pillars that implement it. The overarching objective is to identify the main economic sources of variation in observed Latin American corporate bond spreads and to quantify how these sources evolve across market environments. The framework is directly motivated by the literature showing that bond spreads reflect the joint effect of issuer-specific risk, liquidity conditions, common factors, and broader financial conditions (Elton et al. 2001; Collin-Dufresne, Goldstein, and Martin 2001; Chen, Lesmond, and Wei 2007; Amato and Remolona 2003).

3.1 Three-Level Risk Decomposition Structure

For each bond i at time t, the observed spread $s_{i,t}^{\text{obs}}$ is modelled as a function of three nested information layers:

$$\mathbf{X}_{i,t} = \left(\mathbf{X}_t^{\text{macro}},\ \mathbf{X}_{i,t}^{\text{em}},\ \mathbf{X}_{i,t}^{\text{bond}}\right).$$

- $\mathbf{X}_t^{\text{macro}}$ (**Global block**) collects broad global variables that affect all bonds simultaneously: VIX, MOVE, equity indices (FTSE 100, STOXX 600, S&P 500), credit indices (CDX IG, iTraxx Europe), Bloomberg financial conditions indices, GDP growth, inflation, global liquidity proxies, and geopolitical risk. This block is consistent with the macro-financial interpretation of spread premia developed by Gilchrist and Zakrajšek (2012).

- $\mathbf{X}_{i,t}^{\text{em}}$ (**Emerging / LATAM block**) contains variables capturing the bond's Latin American risk exposure: MSCI EM, JPMorgan EMBI, LATAM CDS and JPM spread composites, LATAM FX direction and volatility scores, EM policy uncertainty, country risk premia, and country-level geopolitical risk for the main LATAM economies. This block captures the regional risk transmission channels specific to Latin America, in line with the evidence that bond pricing in these markets depends jointly on domestic conditions and common regional factors (Miyajima, Mohanty, and Chan 2012; Díaz Weigel and Gemmill 2006; Erb, Harvey, and Viskanta 1999).

- $\mathbf{X}_{i,t}^{\text{bond}}$ (**Bonds block**) contains bond-specific structural and liquidity characteristics: coupon rate, age, issuance size (`SIZE`, `LOG_SIZE`), relative bid–ask spread (`BA_SPREAD`), hard-currency denomination, seniority, sector, and country-of-risk dummies. Raw price levels (`MID_PRICE`, `DIRTY_PRICE`, `BID_PRICE`, `ASK_PRICE`) are *excluded* to prevent a mechanical relationship with the Z-spread target; `BA_SPREAD` is retained as a pure liquidity proxy (Chen, Lesmond, and Wei 2007).

The spread is interpreted as:

$$s_{i,t}^{\text{obs}} = f\left(\mathbf{X}_t^{\text{macro}},\ \mathbf{X}_{i,t}^{\text{em}},\ \mathbf{X}_{i,t}^{\text{bond}}\right) + u_{i,t},$$

and the goal is to recover a decomposition into a **macro component**, an **emerging / LATAM-risk component**, a **bond-specific component**, and a **latent common component** capturing residual co-movement.

3.2 Pillar 1: Data Cleaning and Feature Screening

The first pillar builds the model-ready panel from the raw Bloomberg workbook (`MASTER.xlsx`). It applies sequential universe filters (LATAM ex-Caribbean region, fixed coupons, corporate sector, option-bond exclusion), deduplicates the panel to a **weekly frequency** (last observation per ISIN×ISO-week), and constructs two target variables:

$$\text{MODEL\_TARGET}_{i,t} = \frac{z_{i,t}^{\text{raw}} - z_{i,t-1}^{\text{raw}}}{z_{i,t-1}^{\text{raw}}}, \tag{1}$$

$$\text{MODEL\_TARGET\_LEVEL}_{i,t} = z_{i,t}^{\text{raw}}. \tag{2}$$

`MODEL_TARGET` (weekly percentage change, equation (1)) drives the PCA and HMM stages. `MODEL_TARGET_LEVEL` (spread level, equation (2)) is the dependent variable in the final OLS decomposition. Both variables, together with `Z_SPREAD_BPS`, `Z_SPREAD_MID`, and `TARGET_LAG`, are **hard-blocked from the explanatory feature set** throughout the pipeline.

After universe construction, two sequential screens reduce the candidate feature set to a parsimonious set $\mathcal{S}$:

- **Pairwise correlation screen**: for every pair (j, k) with $|\rho_{jk}| \geq 0.80$, the feature with lower absolute correlation with the target is removed.
- **VIF screen**: features are dropped iteratively in order of decreasing VIF until $\max_j \mathrm{VIF}_j \leq 10$.

Every surviving feature is assigned to a taxonomy block $b(x) \in \{\text{Global, Emerging, Bonds}\}$. The output is `MASTER_clean.xlsx` with two sheets: `clean_data` (the weekly panel) and `selected_features` (the feature metadata).

3.3 Pillar 2: Baseline OLS Block Decomposition and Latent Factor

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ be the standardised feature matrix and $\mathbf{y} \in \mathbb{R}^n$ the vector of spread levels (`MODEL_TARGET_LEVEL`). The baseline model is estimated by OLS with HC3 standard errors:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \alpha\mathbf{1} + \boldsymbol{\varepsilon}.$$

Block-level contributions are:

$$C_{i,t}^{(b)} = \sum_{\substack{x\in\mathcal{S} \\ b(x)=b}} \hat{\beta}_x \, X_{i,t,x}, \qquad b \in \{\text{Global, Emerging, Bonds}\},$$

and the share of explained absolute contribution for block b is:

$$\omega^{(b)} = \frac{\overline{|C^{(b)}|}}{\sum_{b'} \overline{|C^{(b')}|}}.$$

Baseline residuals $\hat{\varepsilon}_{i,t}$ are then pivoted into a $T \times N$ panel $\mathbf{E}$, standardised, and decomposed via PCA: $\mathbf{E} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$. The first principal component score $\widehat{PC}_{1,t} = \mathbf{U}_{:,1}\, D_{1,1}$ is sign-aligned so that $\mathrm{Corr}(\widehat{PC}_{1,t}, \bar{y}_t) \geq 0$. This factor captures residual co-movement not explained by observable covariates, in the spirit of Collin-Dufresne, Goldstein, and Martin (2001) and Díaz Weigel and Gemmill (2006).

3.4 Pillar 3: Regime Identification and Regime-Conditioned Decomposition

A $K = 3$ Gaussian HMM with diagonal covariance is estimated on a period-level matrix $\mathbf{Y}_t$ concatenating $\widehat{PC}_{1,t}$ with macro and LATAM indicators — VIX, MOVE, JPM EMBI, MSCI EM, LATAM CDS credit risk, and LATAM JPM spread risk:

$$\mathbb{P}(R_t = j \mid R_{t-1} = i) = p_{ij}, \tag{3}$$

$$\mathbf{Y}_t \mid (R_t = k) \sim \mathcal{N}\big(\boldsymbol{\mu}_k,\, \mathrm{diag}(\boldsymbol{\sigma}_k^2)\big)\,. \tag{4}$$

The model is re-estimated with $M = 20$ random starts; the highest log-likelihood solution is retained. States are labelled *Calm*, *Transitional*, and *Stress* in ascending order of average $\widehat{PC}_{1,t}$, and a 3-period minimum-duration smoothing pass removes transient flips. Filtered probabilities $\pi_{k,t} = \mathbb{P}(R_t = k \mid \mathbf{Y}_{1:t})$ are joined to the bond-period panel via $\texttt{YEAR\_WEEK}_{i,t}$.

The final decomposition model augments the baseline block scores with latent PC loadings, non-Calm regime probabilities, and block×regime interactions:

$$y_{i,t} = \alpha + \sum_b \gamma_b\, C_{i,t}^{(b)} + \sum_q \delta_q\, \widehat{PC}_{q,t} + \sum_{k\neq \mathrm{C}} \lambda_k\, \pi_{k,t} + \sum_{b,\, k\neq \mathrm{C}} \theta_{b,k}\, C_{i,t}^{(b)}\, \pi_{k,t} + \xi_{i,t}, \tag{5}$$

estimated by OLS with HC3 standard errors. The five risk components are:

$$\begin{aligned}
\hat{C}_{i,t}^{\text{Global}} &= \Big(\hat{\gamma}_{\text{G}} + \textstyle\sum_{k\neq \text{C}} \hat{\theta}_{\text{G},k}\, \pi_{k,t}\Big) C_{i,t}^{(\text{Global})}, \\
\hat{C}_{i,t}^{\text{Emerging}} &= \Big(\hat{\gamma}_{\text{E}} + \textstyle\sum_{k\neq \text{C}} \hat{\theta}_{\text{E},k}\, \pi_{k,t}\Big) C_{i,t}^{(\text{Emerging})}, \\
\hat{C}_{i,t}^{\text{Bonds}} &= \Big(\hat{\gamma}_{\text{B}} + \textstyle\sum_{k\neq \text{C}} \hat{\theta}_{\text{B},k}\, \pi_{k,t}\Big) C_{i,t}^{(\text{Bonds})}, \\
\hat{C}_{i,t}^{\text{Latent}} &= \textstyle\sum_{q} \hat{\delta}_q\, \widehat{PC}_{q,t}, \\
\hat{C}_{i,t}^{\text{Regime}} &= \textstyle\sum_{k\neq \text{C}} \hat{\lambda}_k\, \pi_{k,t}.
\end{aligned}$$

This specification yields a **regime-sensitive risk decomposition** of observed spreads whose relative contributions evolve smoothly across Calm, Transitional, and Stress periods. Three nested models are compared — baseline OLS, baseline augmented by latent PCs, and the full HMM-conditioned decomposition (equation (5)) — using R^2, adjusted R^2, RMSE, and MAE as fit criteria. The use of latent regimes is motivated by Gilchrist and Zakrajšek (2012), Erb, Harvey, and Viskanta (1999), and Wang, Lin, and Mikhelson (2020).

4 Quantitative Framework

This section describes the concrete computational pipeline that implements the three-pillar methodology. It is organized as a two-script chain: `data_cleaning.py` produces a clean, model-ready panel, which is then consumed by `model_risk_decomposition.py` to estimate baseline contributions, latent factors, and regime-conditioned scores. The two final outputs written to disk are `final_monitoring_summary.html` and `row_level_decomposition.csv`, both placed directly in `OUTPUT_DIR` defined by `path_setUp.py`.

4.1 Stage 1: Universe Construction and Data Cleaning

4.1.1 Input and path management

All file paths are resolved through `path_setUp.py`, which defines a single `OUTPUT_DIR` rooted in the network drive. The raw input is `MASTER.xlsx` (stored in `OUTPUT_DIR`); the cleaning output is `MASTER_clean.xlsx` written to the same directory. No intermediate files are produced.

4.1.2 Universe filters

The following filters are applied sequentially to the raw panel:

1. **LATAM ex-Caribbean filter**: rows are retained only when `CNTRY_OF_RISK` belongs to the 17 Latin American economies excluding Caribbean territories. The filter is applied strictly on the *economic* country of risk, not the legal domicile, so offshore issuing vehicles registered in Luxembourg, the Cayman Islands, or the United States are correctly attributed to their issuer's home country.
2. **Fixed-coupon filter**: bonds with `CPN_TYP` $\neq$ `FIXED` are dropped.
3. **Government exclusion**: observations where `INDUSTRY_SECTOR = GOVERNMENT` are removed to restrict the universe to corporate issuers.
4. **Option-bond exclusion** (configurable): callable, putable, and sinkable bonds can be excluded to avoid contaminating Z-spread dynamics with embedded-option premia.

4.1.3 Temporal deduplication

After filtering, the panel is deduplicated to a weekly frequency by retaining the last available observation within each ISIN×ISO-week cell:

$$\text{YEAR\_WEEK}_{i,t} = \text{year}(t)\text{-W}\,\text{week}(t).$$

4.1.4 Derived bond features

The following structural features are computed from raw fields and admitted as explanatory variables:

- **AGE_YEARS**: $(\text{DATE} - \text{ISSUE\_DT})/365.25$, clipped at zero.
- **SIZE**: raw par value (`PAR_VALUE` or `MV`).
- **LOG_SIZE**: $\log(1 + \text{SIZE})$. Captures market depth without price information.
- **BA_SPREAD**: relative bid–ask spread,

$$\text{BA\_SPREAD}_{i,t} = \frac{\text{ASK}_{i,t} - \text{BID}_{i,t}}{0.5\,(\text{ASK}_{i,t} + \text{BID}_{i,t})}.$$

  Serves as a pure liquidity proxy. `BID_PRICE` and `ASK_PRICE` are retained only for this computation and are otherwise hard-blocked from the model feature set.
- **HARD_CCY**: binary indicator for hard-currency denomination.
- **Seniority dummies**: `SENIORITY_SR_UNSECURED`, `SENIORITY_SUBORDINATED`, `SENIORITY_SECURED`, etc., derived from `PAYMENT_RANK` when not already present.

- **Option dummies**: OPTION_CALLABLE, OPTION_PUTABLE, OPTION_SINKABLE, derived from CALLABLE, PUTABLE, SINKABLE.

Price-level fields (MID_PRICE, DIRTY_PRICE, MID_PRICE_CALC) are excluded throughout to prevent a mechanical relationship between bond prices and the Z-spread target.

4.1.5 Categorical dummies

One-hot dummies are generated for CNTRY_OF_RISK, CURRENCY, INDUSTRY_SECTOR, and PAYMENT_RANK, keeping up to 40 levels per variable and collapsing the remainder into OTHER. The first level is dropped to avoid perfect collinearity.

4.1.6 Model-target construction

Two target variables are constructed from the raw Z-spread $z_{i,t}^{\text{raw}}$ (either Z_SPREAD_MID or Z_SPREAD_BPS):

$$\text{MODEL\_TARGET}_{i,t} = \frac{z_{i,t}^{\text{raw}} - z_{i,t-1}^{\text{raw}}}{z_{i,t-1}^{\text{raw}}}, \tag{6}$$

$$\text{MODEL\_TARGET\_LEVEL}_{i,t} = z_{i,t}^{\text{raw}}. \tag{7}$$

MODEL_TARGET (weekly percentage change, equation (6)) is the input to the PCA residual decomposition and the HMM regime identification. MODEL_TARGET_LEVEL (spread level, equation (7)) is the dependent variable $y_{i,t}$ in the final OLS decomposition, so that estimated coefficients are interpretable in basis points. Both variables, together with Z_SPREAD_BPS, Z_SPREAD_MID, and TARGET_LAG, are **hard-blocked from the explanatory feature set** at every stage of the pipeline.

4.2 Stage 2: Feature Screening

After cleaning, p_0 candidate features remain. Two sequential screens reduce this to a parsimonious set $\mathcal{S}$ before any regression is estimated.

4.2.1 Column whitelist

A hard whitelist (ALLOWED_RAW_COLUMNS) defines which columns may enter the pipeline. A second hard-block set (DISALLOWED_MODEL_FEATURES) prevents price levels and target columns from ever appearing as explanatory variables:

$$\texttt{DISALLOWED\_MODEL\_FEATURES} = \underbrace{\{\texttt{MID\_PRICE}, \texttt{DIRTY\_PRICE}, \dots\}}_{\text{price levels}} \cup \underbrace{\{\texttt{Z\_SPREAD\_MID}, \texttt{MODEL\_TARGET}, \dots\}}_{\text{target-only}}.$$

4.2.2 Pairwise correlation screen

For every pair (j, k) with $|\rho_{jk}| \geq \tau_\rho$ (default $\tau_\rho = 0.80$), the feature with the lower absolute correlation with the target is removed:

$$\text{drop } \arg\min_{j \in \{j,k\}} |\operatorname{Corr}(X_j, y)|.$$

4.2.3 VIF screen

Remaining features are standardised and their Variance Inflation Factors are computed:

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is the coefficient of determination from regressing X_j on all other features. Features are dropped iteratively in order of decreasing VIF until $\max_j \text{VIF}_j \leq \tau_{\text{VIF}}$ (default $\tau_{\text{VIF}} = 10$).

4.2.4 Feature taxonomy

Every surviving feature $x \in \mathcal{S}$ is assigned to one of three blocks $b(x) \in \{\text{Global, Emerging, Bonds}\}$ by the `FeatureTaxonomy` classifier. Representative assignments are:

- **Global**: `MA3_VIX_INDEX`, `MA3_SP500`, `MA3_CDX_IG_CDSI_GEN_5Y_CORP`, `MA3_BFCIUS_INDEX`, `MA3_GPR_LEVEL`, ...
- **Emerging**: `MA3_JPM_EMBI`, `MA3_MSCI_EM`, `LATAM_CDS_CREDIT_RISK`, `MA3_GPR_LATAM_MOMENTUM`, `CNTRY_OF_RISK_MX`, ...
- **Bonds**: `COUPON`, `AGE_YEARS`, `SIZE`, `LOG_SIZE`, `BA_SPREAD`, `SENIORITY_SR_UNSECURED`, sector dummies, ...

The cleaned panel and feature metadata are exported as `MASTER_clean.xlsx` with two sheets: `clean_data` and `selected_features`.

4.3 Stage 3: Baseline OLS Block Decomposition

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ be the standardised feature matrix and $\mathbf{y} \in \mathbb{R}^n$ the vector of spread levels (`MODEL_TARGET_LEVEL`). The baseline model is:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \alpha\mathbf{1} + \boldsymbol{\varepsilon},$$

estimated by OLS with HC3 standard errors. Block contributions and shares are:

$$C^{(b)}_{i,t} = \sum_{\substack{x\in\mathcal{S}\\ b(x)=b}} \hat{\beta}_x\, X_{i,t,x}, \qquad \omega^{(b)} = \frac{\overline{|C^{(b)}|}}{\sum_{b'} \overline{|C^{(b')}|}}.$$

4.4 Stage 4: Residual PCA and Latent Common Factor

Baseline residuals $\hat{\varepsilon}_{i,t}$ are pivoted into a $T \times N$ panel $\mathbf{E}$, imputed with each ISIN's median residual, standardised, and decomposed via PCA: $\mathbf{E} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$. The first principal component score $\widehat{PC}_{1,t} = \mathbf{U}_{:,1}\, D_{1,1}$ is sign-aligned so that $\mathrm{Corr}(\widehat{PC}_{1,t},\, \bar{y}_t) \geq 0$. Only periods with at least 5 active ISINs are retained.

4.5 Stage 5: Compact HMM and Regime-Conditioned Decomposition

4.5.1 HMM estimation

A $K = 3$ Gaussian HMM with diagonal covariance is estimated on:

$$\mathbf{Y}_t = \big(\widehat{PC}_{1,t},\ \texttt{MA3\_VIX},\ \texttt{MA3\_MOVE},\ \texttt{MA3\_JPM\_EMBI},\ \texttt{MA3\_MSCI\_EM},\ \texttt{LATAM\_CDS\_CREDIT\_RISK},\ \texttt{LATAM\_JPM\_SPREAD\_CREDIT\_RIS}$$

The model is re-estimated with $M = 20$ random starts; the highest log-likelihood solution is retained. States are labelled *Calm*, *Transitional*, and *Stress* in ascending order of average $\widehat{PC}_{1,t}$, and a 3-period minimum-duration smoothing pass removes transient flips. Filtered probabilities $\pi_{k,t} = \mathbb{P}(R_t = k \mid \mathbf{Y}_{1:t})$ are joined to the bond-period panel via $\texttt{YEAR\_WEEK}_{i,t}$.

4.5.2 Regime-conditioned OLS decomposition

The final model is:

$$y_{i,t} = \alpha + \sum_b \gamma_b\, C^{(b)}_{i,t} + \sum_q \delta_q\, \widehat{PC}_{q,t} + \sum_{k\neq \mathrm{C}} \lambda_k\, \pi_{k,t} + \sum_{b,\, k\neq \mathrm{C}} \theta_{b,k}\, C^{(b)}_{i,t}\, \pi_{k,t} + \xi_{i,t}, \tag{8}$$

estimated by OLS with HC3 standard errors, yielding five risk components ($\hat{C}^{\text{Global}}$, $\hat{C}^{\text{Emerging}}$, $\hat{C}^{\text{Bonds}}$, $\hat{C}^{\text{Latent}}$, $\hat{C}^{\text{Regime}}$) as defined in Section 3.4. Three nested models are compared using R^2, adjusted R^2, RMSE, and MAE.

4.6 Variable Roles and Exclusion Rules

Three hard exclusion rules are enforced at every stage:

- **Target-only variables.** `Z_SPREAD_BPS`, `Z_SPREAD_MID`, `TARGET_LAG`, `MODEL_TARGET`, and `MODEL_TARGET_LEVEL` are removed from the candidate feature set at data-loading time.
- **Price-level exclusion.** `MID_PRICE`, `DIRTY_PRICE`, `BID_PRICE`, `ASK_PRICE`, and `MID_PRICE_CALC` are excluded. `BA_SPREAD` is exempted as a liquidity proxy.
- **Size controls.** `SIZE` and `LOG_SIZE` are retained as structural controls capturing market depth.

Table 1 summarises the role of the main variable groups.

Table 1: Variable roles in the empirical pipeline.

Group	**Examples**	**Role**	**Block**
Spread levels	`Z_SPREAD_MID`, `Z_SPREAD_BPS`	Target only (y)	—
Pct change	`MODEL_TARGET`	HMM / PCA target	—
Level	`MODEL_TARGET_LEVEL`	OLS target (y)	—
Global macro	`MA3_VIX`, `MA3_CDX`, `MA3_SP500`	Explanatory (X)	Global
LATAM / EM	`MA3_JPM_EMBI`, `LATAM_CDS`	Explanatory (X)	Emerging
Bond controls	`COUPON`, `AGE_YEARS`, `LOG_SIZE`	Explanatory (X)	Bonds
Liquidity	`BA_SPREAD`	Explanatory (X)	Bonds
Country dummies	`CNTRY_OF_RISK_MX`, ...	Explanatory (X)	Emerging
Price levels	`MID_PRICE`, `DIRTY_PRICE`	Excluded (bias)	—
Raw bid/ask	`BID_PRICE`, `ASK_PRICE`	`BA_SPREAD` only	—

5 Model Results

This section presents the empirical findings from the three-stage pipeline described in Section 4. Results are organised sequentially: sample characteristics, model fit progression, baseline block decomposition, residual PCA, HMM regime identification, and the final regime-conditioned risk decomposition.

5.1 Sample Overview

Table 2 reports the key sample characteristics and aggregate model-fit statistics for the estimation sample.

Table 2: Sample and model summary statistics.

Metric	**Value**	**Notes**
Observations	7,067	Bond-period rows after all filters
Periods (ISO-weeks)	334	Unique YEAR_WEEK values
ISINs	49	Unique bonds
Model features	50	After correlation and VIF screens
Target mean abs.	918.3608	Weekly pct-change of Z-spread
Target std.	2114.3075	
Baseline OLS R^2	30.7%	Observable blocks only
Baseline Adj. R^2	30.2%	
HMM-cond. R^2	33.1%	Full regime-aware decomposition
HMM-cond. Adj. R^2	32.9%	
PC1 variance share	14.3%	Latent common factor strength
Avg. self-transition	96.4%	Regime persistence

The sample covers a cross-sectional panel of Latin American fixed-coupon corporate bonds observed at weekly frequency, after applying the universe filters described in Section 4.1. The number of active ISINs per period varies over time as bonds enter and exit the index; the feature count reflects the parsimonious set surviving the correlation and VIF screens of Section 4.2.

5.2 Model Fit Progression

Table 3 compares the three nested models on the same estimation sample. Model 1 uses only the observable feature blocks $\mathcal{S}$. Model 2 augments these with the three residual PCA factors. Model 3 is the full HMM-conditioned decomposition (equation (8)). The progressive improvement in R^2 and reduction in RMSE across the three specifications quantifies the incremental contribution of, first, the latent common spread factor and, second, the regime-conditioning layer.

Table 3: Model fit comparison. All three models are estimated on the same in-sample window. Fit metrics measure explanatory power, not out-of-sample forecasting accuracy.

Model	R^2	**Adj.** R^2	**RMSE**	**MAE**	**Augmentation**
1. Baseline OLS	30.7%	30.2%	1760.2017	989.0516	Observable blocks only
2. Baseline + latent PCs	30.7%	30.2%	1759.9941	989.0804	+ Residual PCA factors
3. HMM-conditioned decomposition	33.1%	32.9%	1729.5613	970.5075	+ Regime probs. & interactions

The gap between Model 1 and Model 2 reflects the residual common-factor documented by Collin-Dufresne, Goldstein, and Martin (2001): a large share of co-movement in spread changes is not captured by observable macro-financial variables alone. The further improvement from Model 2 to Model 3 reflects the regime-contingent repricing of all three observable blocks, consistent with the evidence in Gilchrist and Zakrajšek (2012) and Wang, Lin, and Mikhelson (2020) that risk premia vary systematically across market states.

5.3 Baseline OLS Block Decomposition

Table 4 reports the share of total explained absolute spread variation attributable to each of the three observable risk blocks in the baseline OLS model. The share $\omega^{(b)}$ is defined as the time-series and cross-sectional mean of $|C_{i,t}^{(b)}|$ normalised by the sum across all three blocks (Section 3.3).

Table 4: Baseline OLS block contribution shares. $\omega^{(b)} = \overline{|C^{(b)}|} / \sum_{b'} \overline{|C^{(b')}|}$.

Risk block	Share of abs. contribution	Content
Global	14.8%	VIX, MOVE, CDX, FCI, GPR, equity indices
Emerging	37.1%	EMBI, MSCI EM, LATAM CDS, country GPR
Bonds	48.2%	Coupon, age, size, seniority, sector dummies
Total	100.0%	

The Global block encompasses broad macro-financial forces — VIX, MOVE, credit indices, financial conditions, and geopolitical risk — that affect all bonds simultaneously. The Emerging block captures LATAM and EM-specific risk channels, including sovereign spread composites, regional FX dynamics, and country-level geopolitical risk. The Bonds block captures instrument-level structural and liquidity characteristics. The relative magnitudes reflect the degree to which LATAM corporate bond spread *levels* are driven by each layer of the hierarchy described in Section 3.1.

5.4 Residual PCA: Latent Common Spread Factor

Table 5 reports the explained variance of the first three principal components extracted from the cross-sectional OLS residual panel. A high PC1 share is consistent with the strong common factor in spread changes documented by Collin-Dufresne, Goldstein, and Martin (2001) and the regional contagion interpretation of Díaz Weigel and Gemmill (2006).

Table 5: Residual PCA explained variance. The panel is constructed from baseline OLS residuals pivoted across ISINs by ISO-week; periods with fewer than 5 active ISINs are excluded.

Component	Explained variance	Cumulative	Interpretation
PC1	14.3%	14.3%	Latent common spread factor
PC2	7.7%	22.0%	Secondary common component
PC3	6.1%	28.1%	Tertiary common component

The sign of $\widehat{PC}_{1,t}$ is aligned so that $\mathrm{Corr}(\widehat{PC}_{1,t}, \bar{y}_t) \geq 0$: higher PC1 values therefore correspond to periods of elevated spread levels. This latent factor is the primary input to the HMM regime identification described in the next subsection.

5.5 HMM Regime Identification

5.5.1 Regime Duration and Persistence

Table 6 characterises the frequency and persistence of each identified regime. Long mean durations and high diagonal entries in the transition matrix (Table 7) confirm that the three states represent genuine and economically distinct market environments rather than noisy week-to-week fluctuations. This persistence is a necessary condition for the regime-conditioning layer to be economically meaningful, in line with the rationale developed in Wang, Lin, and Mikhelson (2020).

Table 6: HMM regime duration statistics. Duration is measured in ISO-weeks. A 3-period minimum-duration smoothing pass is applied before computing run-length statistics.

Regime	N runs	Mean duration	Median duration	Max duration
Calm	4	36.8	19.5	103
Transitional	5	16.2	4.0	57
Stress	3	35.3	26.0	73

5.5.2 Transition Probabilities

Table 7 reports the estimated one-step transition probabilities $p_{ij} = \mathbb{P}(R_t = j \mid R_{t-1} = i)$. High diagonal entries confirm regime persistence; the off-diagonal structure reveals whether regime shifts tend to pass through the Transitional state or jump directly between Calm and Stress.

Table 7: HMM smoothed-state transition probability matrix. Rows = state at $t-1$; columns = state at t.

	→ Calm	→ Transitional	→ Stress
Calm	97.3%	2.0%	0.7%
Transitional	4.9%	93.8%	1.2%
Stress	0.0%	1.9%	98.1%

5.6 Final HMM-Conditioned Risk Decomposition

5.6.1 Overall Component Shares

Table 8 reports the share of total explained absolute spread variation attributable to each of the five components in the final HMM-conditioned model (equation (8)). Comparing these shares with the baseline shares in Table 4 reveals the incremental contribution of the latent common factor and the regime-probability channel.

Table 8: Final HMM-conditioned decomposition — overall component shares. $\omega^{(c)} = |\hat{C}^{(c)}| / \sum_{c'} |\hat{C}^{(c')}|$.

Component	Share of abs. contribution	Description
Bonds	44.7%	Bond-level structural controls
Emerging	34.9%	LATAM and EM risk factors
Global	14.4%	Global macro and financial conditions
Regime	4.8%	HMM regime probability loadings
Latent	1.0%	Common residual repricing ($\widehat{PC}_1$)
Total	100.0%	

The Latent component captures the common residual repricing force $\widehat{PC}_{1,t}$ that persists after controlling for all three observable blocks. The Regime component reflects the direct effect of elevated Transitional and Stress probabilities on spread levels, independently of the block contributions. Together, these two components measure the portion of spread variation that observable macro and bond variables cannot account for.

5.6.2 Risk Contributions by Regime

Table 9 decomposes the mean absolute contribution of each risk component conditional on the identified market regime. Differences across rows reveal how the balance between global, emerging, bond-level, latent, and regime forces shifts between Calm, Transitional, and Stress periods. This table constitutes the central empirical result of the thesis: it shows which risk channels become dominant as market conditions deteriorate, providing a direct quantification of the regime-contingent spread decomposition derived in Section 3.4.

Table 9: Mean absolute risk contribution by HMM regime and component. Values are in the same units as the decomposition target (`MODEL_TARGET_LEVEL`, Z-spread in basis points). Regime labels are assigned in ascending order of average $\widehat{PC}_{1,t}$.

Regime	Global	Emerging	Bonds	Latent	Regime
Calm	336.0541	740.2683	1096.5535	13.8697	1.0889
Transitional	288.7798	584.1199	899.9416	26.8776	112.4902
Stress	164.4161	506.7681	479.0432	12.7323	97.8206

The cross-regime comparison is directly motivated by the evidence in Gilchrist and Zakrajšek (2012) that the excess bond premium rises sharply in stress periods, and by Erb, Harvey, and Viskanta (1999) and Díaz Weigel and Gemmill (2006) showing that common and external factors dominate EM spread dynamics during crises. In the Latin American context, the Emerging and Latent components are expected to amplify materially in Stress regimes, reflecting the regional contagion and capital-flow reversal dynamics documented in the literature.

5.6.3 Top Model Coefficients

Table 10 lists the 20 features with the largest absolute OLS coefficients in the baseline model, together with their block attribution. Because all features are standardised before estimation, coefficients are directly comparable across variables and measure the contribution to Z-spread levels in basis points per unit of standardised input.

Table 10: Top-20 baseline OLS coefficients ranked by absolute magnitude. Features are standardised; coefficients are in basis points of Z-spread level per unit of standardised regressor.

Feature	Block	Coefficient	|Coeff.|
BA_SPREAD	Bonds	795.6630	795.6630
SENIORITY_1ST_LIEN	Bonds	-778.3025	778.3025
COUPON	Bonds	-517.4720	517.4720
ISSUE_YEAR	Bonds	-507.5158	507.5158
INDUSTRY_SECTOR_UTILITIES	Bonds	463.6369	463.6369
PAYMENT_RANK_SECURED	Bonds	-374.1014	374.1014
INDUSTRY_SECTOR_FINANCIAL	Bonds	-326.9234	326.9234
INDUSTRY_SECTOR_INDUSTRIAL	Bonds	309.1255	309.1255
MA3_EPUCEMEC_INDEX	Emerging	304.4554	304.4554
LOG_SIZE	Bonds	-274.9548	274.9548
MA3_GPR_CHL_VOL	Emerging	-273.7438	273.7438
INDUSTRY_SECTOR_CONSUMER, NON_CYCLICAL	Bonds	-257.6995	257.6995
MA3_CRPTMEXI_INDEX	Emerging	-253.6972	253.6972
MA3_GPR_ACTS_SHARE	Global	-224.3574	224.3574
OPTION_CALLABLE	Bonds	197.6330	197.6330
CNTRY_OF_RISK_PE	Emerging	-182.3295	182.3295
MA3_CAFZUUEM_INDEX	Emerging	-176.6548	176.6548
MA3_GPR_BRA_MOMENTUM	Emerging	175.4754	175.4754
MA3_GPR_COL_MOMENTUM	Emerging	-174.3526	174.3526
CNTRY_OF_RISK_CL	Emerging	-172.1835	172.1835

A Variable Descriptions

All standard variables are sourced from Bloomberg. Macro and EM variables enter as 3-month moving averages (`MA3_` prefix). Type codes: B = Bloomberg raw; D = derived; Alt = alternative data.

Table 11: Bond-level variables.

Variable	T	Description
`Z_SPREAD_MID`	B	Z-spread: constant z s.t. $P = \sum_t CF_t/(1 + r_t + z)^t$. Primary target.
`CPN_TYP/COUPON`	B	Coupon type and rate. Fixed-coupon bonds only.
`PAR_VALUE`	B	Nominal par value. Basis for `SIZE`.
`MATURITY/ISSUE_DT`	B	Maturity and issue dates. Basis for `AGE_YEARS`.
`PAYMENT_RANK`	B	Seniority (Sr. Unsecured, Sub., 1st/2nd Lien, Secured). Binary dummies.
`CALLABLE/PUTABLE/SINKABLE`	B	Option flags. Optionally excluded to avoid Z-spread contamination.
`INDUSTRY_SECTOR`	B	Bloomberg Level 1 sector. One-hot encoded; government excluded.
`CNTRY_OF_RISK`	B	ISO2 economic country of risk. LATAM filter and country dummies.
`CURRENCY`	B	Denomination currency. Currency dummies.
`AGE_YEARS`	D	(DATE − ISSUE_DT)/365.25. Seasoning proxy.
`SIZE/LOG_SIZE`	D	Par value and log(1+`SIZE`). Market-depth controls; no price info.
`BA_SPREAD`	D	(ASK−BID)/[0.5(ASK+BID)]. Liquidity proxy (Chen, Lesmond, and Wei 2007).
`HARD_CCY`	D	Binary: hard-currency denomination (USD, EUR, GBP, JPY, ...).

Table 12: Global macro variables (Bloomberg, `MA3_` prefix).

Variable	Description
`MA3_VIX_INDEX`	CBOE VIX. 30-day S&P 500 implied vol. Global risk gauge.
`MA3_MOVE_INDEX`	MOVE index. U.S. Treasury implied vol. Rate uncertainty.
`MA3_UKX` / `STOXX600` / `SP500`	FTSE 100, STOXX 600, S&P 500 total returns. DM equity conditions.
`MA3_CDX_IG_...._5Y`	CDX NA IG 5Y CDS. U.S. IG credit benchmark.
`MA3_ITRX_EUR_...._5Y`	iTraxx Europe 5Y CDS. European IG credit counterpart.
`MA3_BFCIUS` / `BFCIEU`	Bloomberg U.S. and European Financial Conditions Indices.
`MA3_GDP_CQOQ_INDEX`	GDP growth (QoQ). Global business cycle.
`MA3_CPI_YOY_INDEX`	CPI inflation (YoY). Global price pressure.
`MA3_EUGNEMUQ_INDEX`	Euro-area unemployment. Labour market slack.
`MA3_ECCPEMUY_INDEX`	Euro-area CPI (YoY). European inflation signal.

Table 13: Emerging and LATAM variables.

Variable	T	Description
`MA3_MSCI_EM`	B	MSCI EM total return. Broad EM equity sentiment.
`MA3_JPM_EMBI`	B	JPMorgan EMBI spread. Sovereign USD spread benchmark.
`LATAM_CDS_CREDIT_RISK`	D	GDP-weighted composite of 5Y sovereign CDS (BR, MX, CO, CL, PE).
`LATAM_JPM_SPREAD`	D	GDP-weighted composite of country EMBI spreads, main LATAM economies.
`MA3_LATAM_FX_DIRECTION`	D	GDP-weighted FX direction for BRL, COP, CLP, PEN, MXN vs. USD.
`MA3_LATAM_FX_VOL`	D	GDP-weighted LATAM FX volatility. External financing stress.
`MA3_CRPT*`	Alt	Bloomberg Country Risk Premia for BR, MX, CL, CO, PE, CR. Blend sovereign CDS, macro fundamentals, and governance.
`MA3_EPUCEMEC_INDEX`	Alt	EM Policy Uncertainty (Baker, Bloom, and Davis 2016). Newspaper-based, GDP-weighted across EM economies.
`MA3_CAFZUUEM_INDEX`	Alt	EM Financial Conditions Index. Rates, credit, equity, FX, flows.
`MA3_AONILS_INDEX`	Alt	Aon Global Liquidity Score. CB and interbank data composite.
`MA3_GLCR*` / `GCIL*`	Alt	Global credit cycle ratios and credit impulse flow indices.
`MA3_GPR*`	Alt	Geopolitical Risk (Caldara and Iacoviello 2022). Text-based; global, LATAM regional, and country level (BR, CL, CO, MX, PE, CR) as level, momentum, and volatility.

A.1 Alternative Data

A.1.1 EM Economic Policy Uncertainty — EPUCEMEC (Baker, Bloom, and Davis 2016)

Text-based index constructed from newspaper article counts mentioning *economy*, *uncertainty*, and *policy* across major EM economies, aggregated with GDP weights. Captures policy ambiguity independently of market prices.

A.1.2 EM Financial Conditions Index — CAFZUUEM

Research-grade composite blending interest rate spreads, credit growth, equity valuations, exchange rate pressure, and portfolio flows into a single standardised EM financial-tightness score. Not a traded instrument.

A.1.3 Global Liquidity and Credit Impulse Indices — AONILS, GLCR*, GCIL*

`AONILS` (Aon Global Liquidity Score) measures global liquidity availability from central bank and interbank data. `GLCR*` indices capture the global credit cycle via private credit-to-GDP, central bank assets, and non-central-bank credit ratios. `GCIL*` indices measure the *flow* of new credit extension relative to GDP (manufacturing and natural-resources channels), based on bank lending surveys and monetary aggregates.

A.1.4 Geopolitical Risk Indices — GPR* (Caldara and Iacoviello 2022)

Text-based indices constructed from ten major international newspapers, counting articles on geopolitical threats and acts normalised by total coverage. Three aggregation levels are used: (i) global (`MA3_GPR_LEVEL`, `_MOMENTUM`, `_VOL`, `_ACTS_SHARE`); (ii) LATAM regional (`MA3_GPR_LATAM_*`); and (iii) country-level series for Brazil, Chile, Colombia, Mexico, Peru, and Costa Rica (`MA3_GPR_BRA_*`, `MA3_GPR_CHL_*`, etc.), each available as level, momentum, and volatility. Purely text-based; independent of market prices or CDS.

References

Amato, Jeffery D. and Eli M. Remolona (Dec. 2003). "The Credit Spread Puzzle". In: *BIS Quarterly Review*. Bank for International Settlements. Commonly circulated as a BIS QRT article; PDF is often hosted under BIS QRT archives (file name r_qt0312e.pdf).

Baker, Scott R., Nicholas Bloom, and Steven J. Davis (2016). "Measuring Economic Policy Uncertainty". In: *The Quarterly Journal of Economics* 131.4, pp. 1593–1636. DOI: 10.1093/qje/qjw024. URL: https://doi.org/10.1093/qje/qjw024.

Caldara, Dario and Matteo Iacoviello (2022). "Measuring Geopolitical Risk". In: *American Economic Review* 112.4, pp. 1194–1225. DOI: 10.1257/aer.20191823. URL: https://doi.org/10.1257/aer.20191823.

Chen, Long, David A. Lesmond, and Jason Wei (2007). "Corporate Yield Spreads and Bond Liquidity". In: *The Journal of Finance* 62.1, pp. 119–149.

Collin-Dufresne, Pierre, Robert S. Goldstein, and J. Spencer Martin (2001). "The Determinants of Credit Spread Changes". In: *The Journal of Finance* 56.6, pp. 2177–2207.

Díaz Weigel, Diana and Gordon Gemmill (2006). "What Drives Credit Risk in Emerging Markets? The Roles of Country Fundamentals and Market Co-movements". In: *Journal of International Money and Finance* 25.3, pp. 476–502.

Elton, Edwin J. et al. (2001). "Explaining the Rate Spread on Corporate Bonds". In: *The Journal of Finance* 56.1. Often circulated earlier as a working paper; published in *The Journal of Finance* (2001)., pp. 247–277.

Erb, Claude B., Campbell R. Harvey, and Tadas E. Viskanta (1999). "Understanding Emerging Market Bonds". In: Draft manuscript, October 21, 1999.

Feldhütter, Peter and Stephen Schaefer (2018). "The Myth of the Credit Spread Puzzle". In: *Review of Financial Studies* 31.8, pp. 2897–2942.

Gilchrist, Simon and Egon Zakrajšek (2012). "Credit Spreads and Business Cycle Fluctuations". In: *American Economic Review* 102.4, pp. 1692–1720.

Jacobs, Michael and Ahmet K. Karagozoglu (2010). *Measuring Credit Risk: CDS Spreads vs. Credit Ratings*. Tech. rep. Working paper version commonly cited in practice; journal metadata varies across circulating versions. SSRN Working Paper.

Miyajima, Ken, M. S. Mohanty, and Tracy Chan (Nov. 2012). *Emerging Market Local Currency Bonds: Diversification and Stability*. Tech. rep. 391. Monetary and Economic Department. Bank for International Settlements.

Wang, Matthew, Yi-Hong Lin, and Ilya Mikhelson (2020). "Regime-Switching Factor Investing with Hidden Markov Models". In: *Journal of Investment Strategies* 9.4, pp. 1–29.